

Subject : Digital Logic

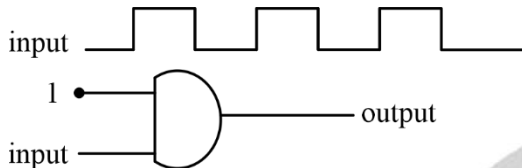
Chapter : Logic Gates

Practice Sheet - 1

[MCQ]

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1. The input wave form is given,



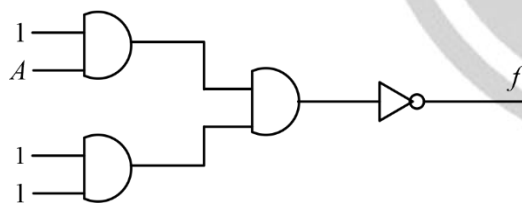
Draw output wave form.

- (a)
- (b) 0
- (c) 1
- (d)

[MCQ]

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2.



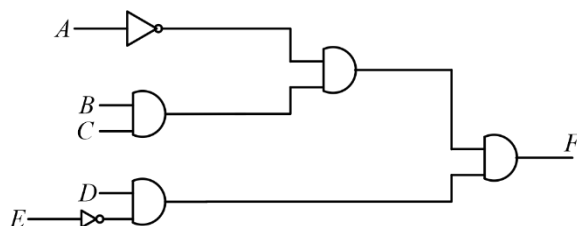
Find value of f .

- (a) 1 (b) 0
- (c) A (d) $\bar{A}$

[MCQ]



3. Find expression of F .



- (a) $\bar{A}BC + D\bar{E}$ (b) $\bar{A}BCDE$
- (c) $(\bar{A} + BC)D\bar{E}$ (d) $\bar{A}BC(D + \bar{E})$

[NAT]

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4. Determine the total number of possible input combinations for a 4-input AND gate.

[MCQ]

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5. The following specifies the way a logic circuit is to operate to initiate seat belt warning in a car “if the driver is present AND the driver NOT buckled up AND the ignition switch is on, then initiate seat belt warning”.

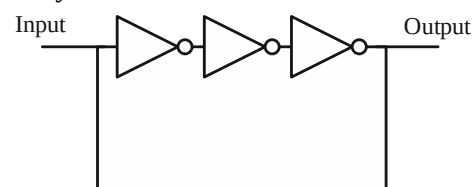
Let logic ‘x’ be 1 for driver present and 0 for driver absent, logic ‘y’ be 1 for seat buckled and 0 for seat NOT buckled, logic ‘z’ be 1 for ignition switch on and 0 for ignition switch off. Let logic ‘w’ be 1 for warning initiated and 0 for no warning. Then, the correct logic to initiate warning, is

- (a)
- (b)
- (c)
- (d)

[NAT]

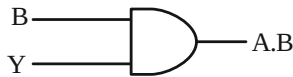
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6. Propagation delay of NOT gate is 1nsec, find the output frequency _____ MHz



[MCQ]

7. A logical circuit is as shown below,



which of the following circuit can be used to get the desired expression?

- (a)
- (b)
- (c)
- (d)

[MCQ]

8. For the given truth table

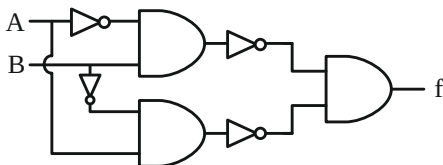
A	B	f
0	0	0
0	1	1
1	0	1
1	1	1

select the respective combination diagram.

- (a)
- (b)
- (c)
- (d)

[MCQ]

9.



Which of the following is true?

- (a) Coincidence detector
- (b) Anti-coincidence detector
- (c) All zero-input detector
- (d) All one input detector

[MCQ]

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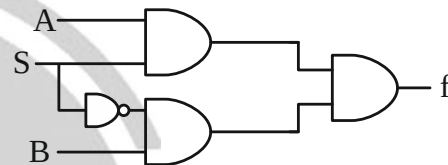
10. Which of the following is not an inverter?

- (a)
- (b)
- (c)
- (d)

[MCQ]

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11.



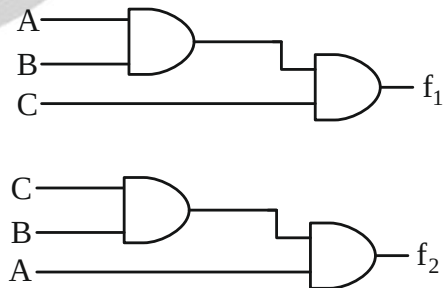
$f = 1$ for combination of A, B, S

- (a) $A = 0, B = 0, S = 0$
- (b) $A = 0, B = 1, S = 0$
- (c) $A = 1, B = 1, S = 0$
- (d) None

[MCQ]

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12.

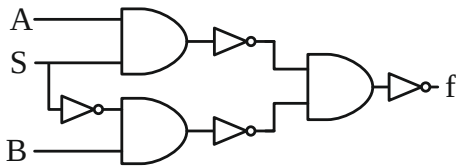


- (a) $f_1 = f_2$ for $A = 1, B = 0, C = 1$
- (b) $f_1 = f_2$ for $A = 1, B = 1, C = 0$
- (c) f_1 can never be equal to f_2
- (d) f_1 always equal to f_2

[MCQ]

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13.

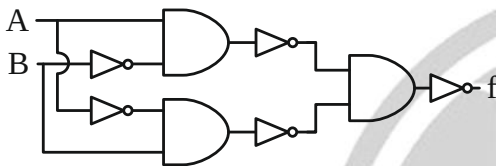
Output at $S = 0$ is

- (a) A (b) B
(c) $\overline{A \cdot B}$ (d) $\overline{AB}$

[MCQ]

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14.



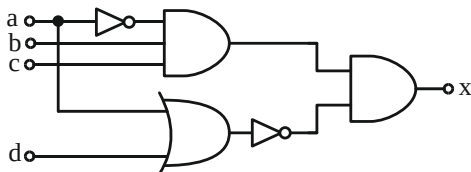
f is an

- (a) Equality detector
(b) Inequality detector
(c) All zero-input detector
(d) All one input detector

[MCQ]

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15. In the figure shown below, if each AND gate is changed to an OR gate and OR gate to an AND gate then the expression for output x will be,



- (a) $x = \overline{a} + \overline{b} + c + \overline{d}$
(b) $x = \overline{a} + b + c + \overline{d}$
(c) $x = a + \overline{b} + \overline{c} + d$
(d) $x = a + \overline{b} + c + d$

A combinational digital circuit design is attempted using only NAND logic. The circuit has four inputs A, B, C, D and one output Z. The output Z is required to be 1 (high) whenever either $A = B = 1$ or $C = D = 1$ or both.

[MCQ]

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16. The minimum number of NAND gates required, will be

- (a) 2 (b) 3
(c) 4 (d) 5

A combinational digital circuit design is attempted using only NAND logic. The circuit has four inputs A, B, C, D and one output Z. The output Z is required to be 1 (high) whenever either $A = B = 1$ or $C = D = 1$ or both.

[MCQ]



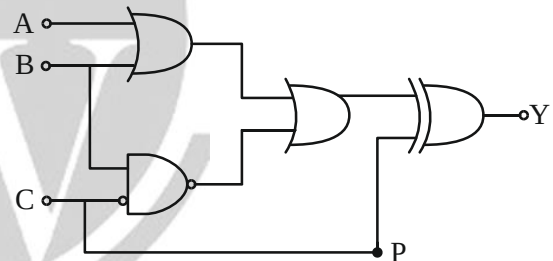
17. If each NAND gate has delay of 10 ns, then on applying inputs the output will appear after

- (a) 10 ns (b) 20 ns
(c) 30 ns (d) 40 ns

[MCQ]

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18. What will be the output Y of combinational digital circuit shown below, if P is stuck at 1?

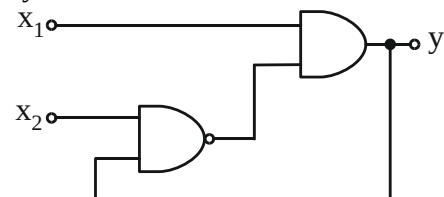


- (a) $A + B + C$ (b) $\overline{A}\overline{B}\overline{C}$
(c) $\overline{ABC}$ (d) 0

[MCQ]

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19. In the circuit shown below, as long as $x_1 = x_2 = 1$, the output y remains

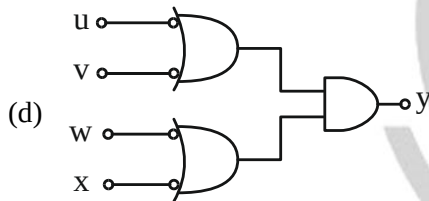
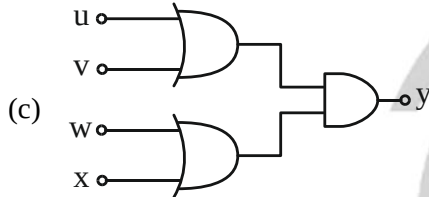
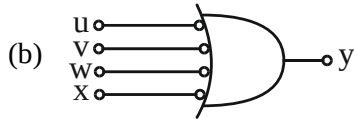
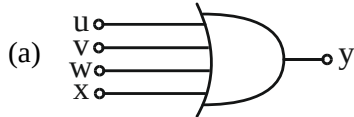
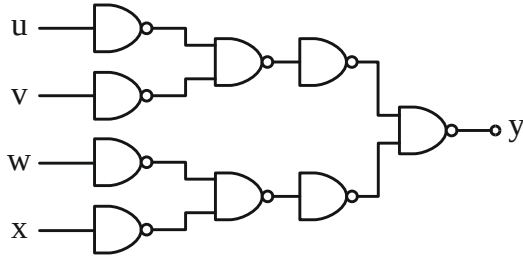


- (a) at 0 (b) at 1
(c) at its initial value (d) unstable

[MCQ]



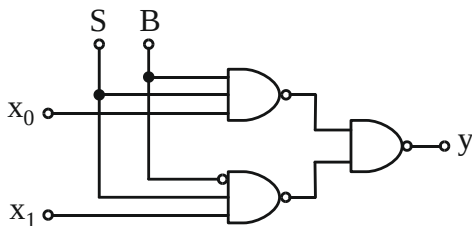
20. The logic circuit shown below, is equivalent to



[MCQ]



21. A logic circuit and its truth table are shown below.



S	B	y
0	0	p
0	1	q
1	0	r
1	1	s

Choose the correct option,

- (a) $p = q = 0, r = x_1, s = x_0$
 (b) $p = q = 1, r = x_0, s = x_1$
 (c) $p = 1, q = 0, r = 0, s = x_1$
 (d) $p = 0, q = 1, r = x_0, s = x_1$

[MCQ]



22. The number count of 2-input NOR gates required to realize $y_1 = \overline{x_1 + x_2 + x_3 + x_4}$ is n_1 and the number count of 2-input NAND gates required to realize $y_2 = \overline{x_1 x_2 x_3 x_4}$ is n_2 . Then

- (a) $n_1 = n_2 = 5$ (b) $n_1 = n_2 = 4$
 (c) $n_1 = n_2 = 6$ (d) $n_1 = n_2 = 3$

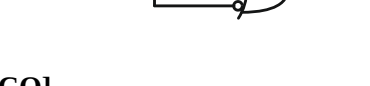
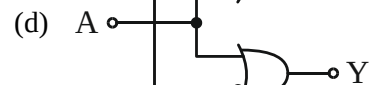
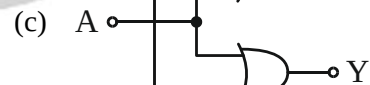
[MCQ]



23. A pulse steering circuit with input signal A, control input B and outputs X and Y, is to be designed to operate as follows:

- (i) When $B = 1$, output X is to follow input A and output $Y = 0$
 (ii) When $B = 0$, output $X = 0$ and output Y is to follow input A.

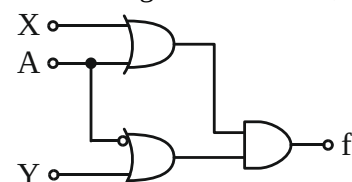
Which one of the following will meet the design goal?



[MCQ]



24. The circuit shown below implements $f(A, B) = A + B$. The values to be assigned to X and Y, will be



- (a) $X = 0, Y = B$ (b) $X = A, Y = B$
 (c) $X = B, Y = 1$ (d) $X = \bar{B}, Y = 0$

[MCQ]

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25. The minimum number of 2 input NAND gates required to implement the following Boolean function $f = (\bar{x} + \bar{y})(z + w)$ is

- (a) 3 (b) 4
(c) 5 (d) 6

[MCQ]

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26. A circuit which is working as NAND gate with positive level logic system will work as _____ gate with negative level logic system.

- (a) NAND (b) NOR
(c) AND (d) OR

[MCQ]

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27. In a two-input NAND gate, if both inputs are shorted, it will behave like a _____ gate.

- (a) Buffer
(b) AND
(c) NOT
(d) EX-OR

[MCQ]

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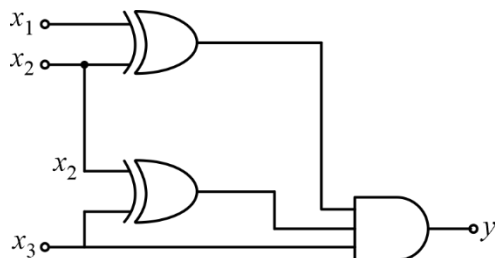
28. The minimum number of NAND gates required to implement the expression $((A + B)C)D$ is

- (a) 6
(b) 5
(c) 8
(d) 4

[MCQ]

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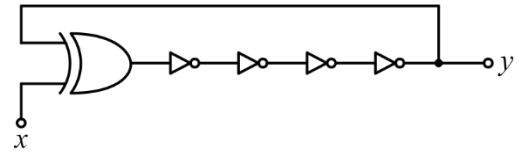
29. The digital circuit shown below, is observed to generate output $y = 1$. Which one of the following input conditions does so?



- (a) $x_1 = x_3 = 1, x_2 = 0$
(b) $x_1 = 0, x_2 = x_3 = 1$
(c) $x_1 = x_2 = x_3 = 0$
(d) $x_1 = x_2 = x_3 = 1$

Common data for Q.No.-30 & 31

All the logic gates in the circuit shown below, have equal finite propagation delay.



[MCQ]

30. The circuit can be used as clock generator, if ★★☆☆

- (a) $x = 0$
(b) $x = 1$
(c) $x = 0$ or 1
(d) $x = y$

[MCQ]

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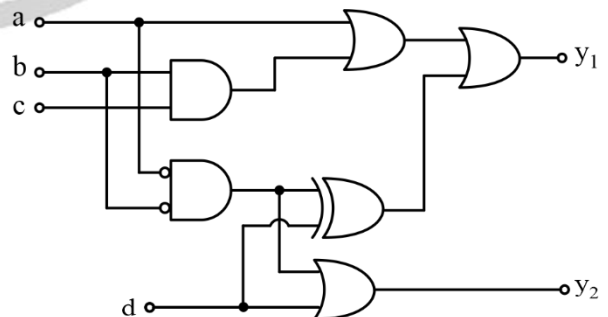
31. If the output waveform has frequency of 10 MHz, the propagation delay of each logic gate, is

- (a) 5 ns
(b) 10 ns
(c) 20 ns
(d) 50 ns

[MCQ]

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32. In the circuit shown below, the input b is stuck at '0' fault. Then,



- (a) $y_1 + y_2 = a \oplus d$ (b) $y_1 + y_2 = 0$
(c) $y_1 y_2 = 1$ (d) $y_1 y_2 = \overline{a \oplus d}$

[NAT]

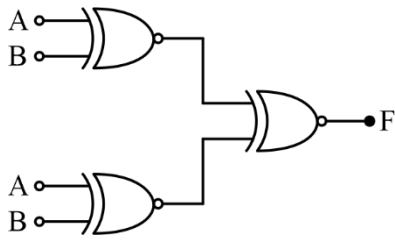
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33. Let $1010 \oplus B = 1100$ where $\oplus$ denotes X-OR operation performed bit wise. The value of B in decimal is _____.

[MCQ]



34. The output of the circuit shown in figure is equal to

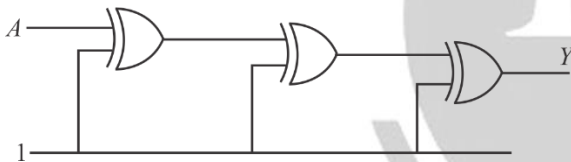


- (a) 0
 (b) 1
 (c) $\bar{A}B + A\bar{B}$
 (d) $\overline{(A \odot B)} \odot \overline{(A \odot B)}$

[MCQ]



35. The initial output of the following circuit is 1. If we apply 010101 at input A (first bit is zero), then what is the bit pattern generated at the output Y.

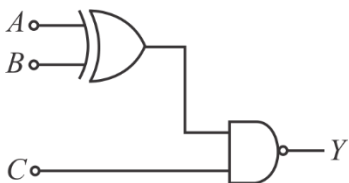


- (a) 010101 (b) 101010
 (c) remains at 0 (d) remains at '1'

[MCQ]



36. The Boolean expression of the output of the logic circuit shown in figure is

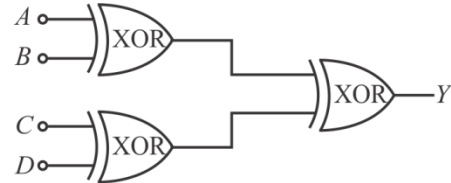


- (a) $Y = \bar{A}\bar{B} + AB + \bar{C}$
 (b) $Y = \bar{A}\bar{B} + AB + C$
 (c) $Y = \bar{A}B + A\bar{B} + C$
 (d) $Y = \bar{A}B + \bar{A}\bar{B} + C$

[MCQ]



37. A, B, C and D are input, and Y is the output bit in the XOR gate circuit of the figure below. Which of the following statements about the sum S of A, B, C, D and Y is correct?

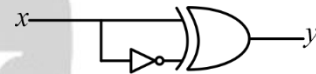


- (a) S is always either zero or odd.
 (b) S is always either zero or even.
 (c) $S = 1$ only if the sum of A, B, C and D is even.
 (d) $S = 1$ only if the sum of A, B, C and D is odd.

[MCQ]



38. The output Y of the logic circuit given below is

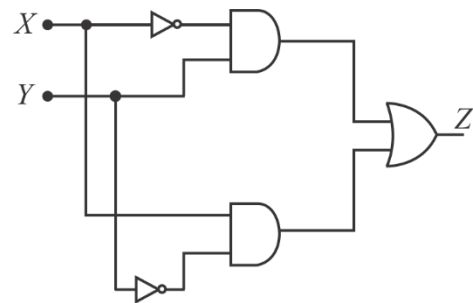


- (a) 1 (b) 0
 (c) x (d) $\bar{x}$

[MCQ]



39. In the circuit shown below, X and Y are digital inputs, and Z is a digital output. The equivalent circuit is



- (a) XNOR gate (b) NOR gate
 (c) NAND gate (d) XOR gate

[MCQ]



40. Which one of the following gates is also known as equivalence gate?

- (a) EX-OR (b) AND
 (c) EX-NOR (d) NOR

[MCQ]

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41. The logic function $f = \overline{xy + \bar{x}\bar{y}}$ is equal to

- (a) EX-NOR (b) NAND
(c) EX-OR (d) NOR

[NAT]

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42. The minimum number of 2 input NOR gates required to implement a 2 input XOR gate is_____.

[MCQ]

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43. The minimum number of 2 input NAND gates required to implement a 2 input EX-NOR gate is

- (a) 4 (b) 5
(c) 3 (d) 6



Answer Key

- | | |
|---|--|
| <p>1. (d)</p> <p>2. (d)</p> <p>3. (b)</p> <p>4. (16)</p> <p>5. (c)</p> <p>6. (166.6)</p> <p>7. (d)</p> <p>8. (c)</p> <p>9. (a)</p> <p>10. (c)</p> <p>11. (d)</p> <p>12. (d)</p> <p>13. (b)</p> <p>14. (b)</p> <p>15. (b)</p> <p>16. (b)</p> <p>17. (b)</p> <p>18. (d)</p> <p>19. (d)</p> <p>20. (a)</p> <p>21. (a)</p> <p>22. (a)</p> | <p>23. (a)</p> <p>24. (c)</p> <p>25. (b)</p> <p>26. (b)</p> <p>27. (c)</p> <p>28. (b)</p> <p>29. (a)</p> <p>30. (b)</p> <p>31. (b)</p> <p>32. (d)</p> <p>33. (6)</p> <p>34. (b)</p> <p>35. (b)</p> <p>36. (a)</p> <p>37. (b)</p> <p>38. (a)</p> <p>39. (d)</p> <p>40. (c)</p> <p>41. (c)</p> <p>42. (5)</p> <p>43. (b)</p> |
|---|--|



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